MODULE 4.3

Review questions LS

Remembering

1 Define the terms:
   a comparative anatomy
   b transitional fossil
   c homologous structure.

2 What term best describes each of the following?
   a the study of the development, structure and function of embryos
   b all the places on Earth where a particular species is found
   c the study of the molecules and chemical reactions found in living organisms
   d characteristics shared by two species that look similar but do not share a common ancestry.

3 List homologous structures in fish and bird embryos that have been shown to be affected by the gene called Gcm-2.

4 What changes occurred in horses in the geological record?

5 What fossil group are birds most closely related to and what features do they share?

Understanding

6 Flying foxes and birds have wings that enable them to fly. Are the wings of these two different organisms homologous structures or analogous structures? Explain.

7 Explain what is meant by *genetic isolation*.

8 Discuss how the fossil record supports the theory of evolution.

9 Explain why a barrier is necessary before speciation can occur.

10 Why do biologists say that species with the same anatomy are related?

Applying

11 How are the following two facts linked? The Queensland lungfish is a lobe-finned fish. The same gene controls development of fingers in humans and fin digits in the lungfish.

12 Use the model of speciation to explain why isolated islands often have a high proportion of species not found elsewhere, even on neighbouring islands.

13 Some early lobe-finned fish are difficult to classify into fish or amphibians, because they have features of both. How does this support the theory of evolution?

14 How are the following facts linked? Whales have some bones that are homologous with the pelvis and rear limb of living tetrapods but do not function as limbs. Whales breathe air using lungs. Whale fossils are always younger than fossils of the oldest land mammals.

Analysing

15 Between 1973 and 2006, scientists studied the ground finch (*Geospiza fortis*) on the tiny Galapagos island of Daphne. The ground finch is a small seed-eating finch with a body mass of only 20 grams (Figure 4.3.14). In 1976, there was a severe drought. By the end of 1978, the average size of the finch's beak had increased.



FIGURE 4.3.14 The ground finch, *Geospiza fortis*, feeds on seeds.

Figure 4.3.15 on page 166 shows what happened to the population of finches, food supply and the hardness and size of the seeds available.